Day 78 - Azure Key Vault

Objective

To understand and implement Azure Key Vault for securely storing secrets and keys, and to use them in an application.

Steps Performed

Created Azure Key Vault (TusharVault) in Azure Portal.

Configured access policies and permissions.

Created a Secret inside Key Vault.

Created a Key inside Key Vault.

Accessed Key Vault using .NET Console Application.

Used DefaultAzureCredential for authentication.

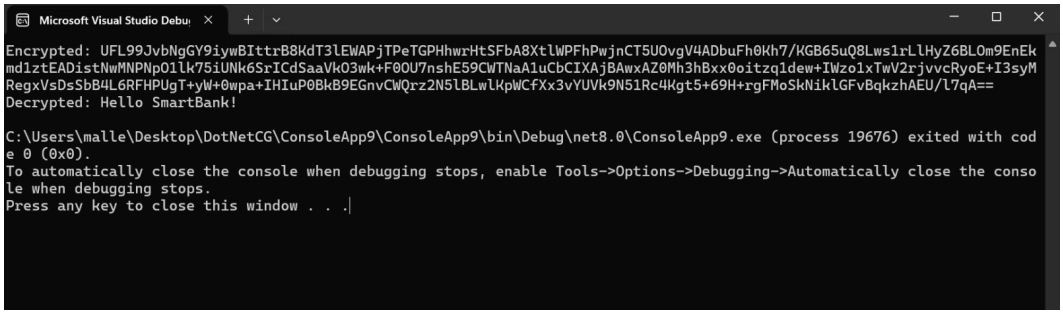
Implemented encryption and decryption using Azure Key Vault keys.

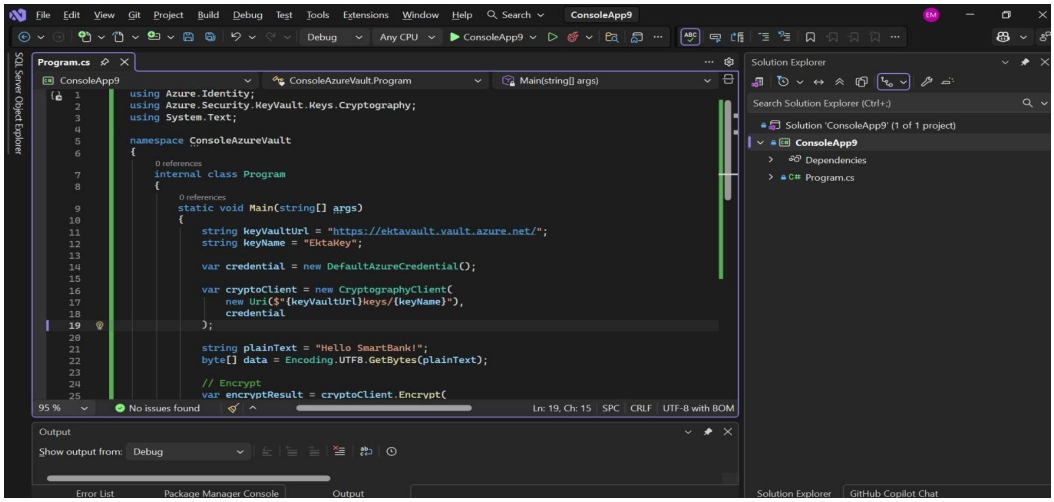
Verified output in console.

Implementation (Code Summary)

Used Azure.Security.KeyVault.Keys.Cryptography with DefaultAzureCredential to perform encryption and decryption.

Screenshots




File Edit View Git Project Build Debug Test Tools Extensions Window Help Search ConsoleApp9
Debug Any CPU ConsoleApp9
Program.cs
ConsoleApp9 ConsoleAzureVault.Program Main(string[] args)
using Azure.Identity;
using Azure.Security.KeyVault.Keys.Cryptography;
using System.Text;
namespace ConsoleAzureVault
{
0 references
internal class Program
{
0 references
static void Main(string[] args)
{
string keyVaultUrl = "https://ektavault.vault.azure.net/";
string keyName = "EktaKey";
var credential = new DefaultAzureCredential();
var cryptoClient = new CryptographyClient(
new Uri($"{keyVaultUrl}keys/{keyName}"),
credential
);
string plainText = "Hello SmartBank!";
byte[] data = Encoding.UTF8.GetBytes(plainText);
// Encrypt
var encryptResult = cryptoClient.Encrypt(
95 % No issues found Ln: 19, Ch: 15 SPC CRLF UTF-8 with BOM
Output
Show output from: Debug
Error List Package Manager Console Output
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'ConsoleApp9' (1 of 1 project)
ConsoleApp9
Dependencies
Program.cs
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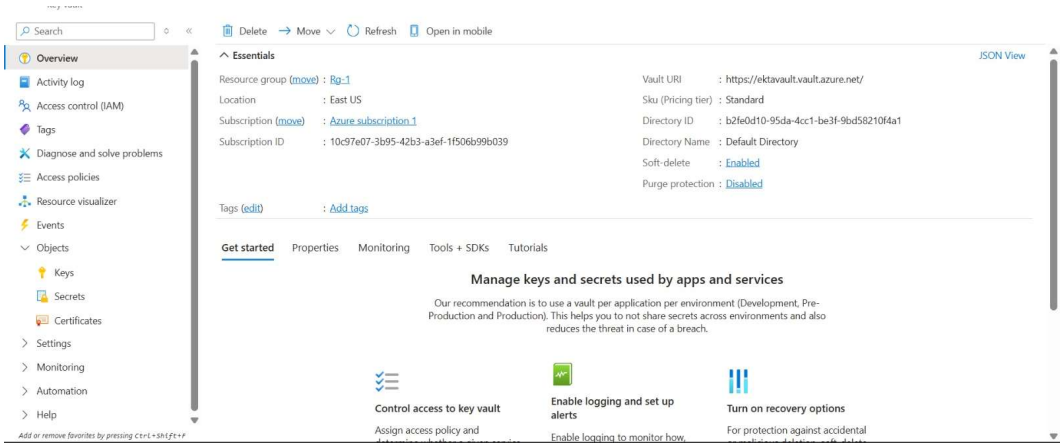
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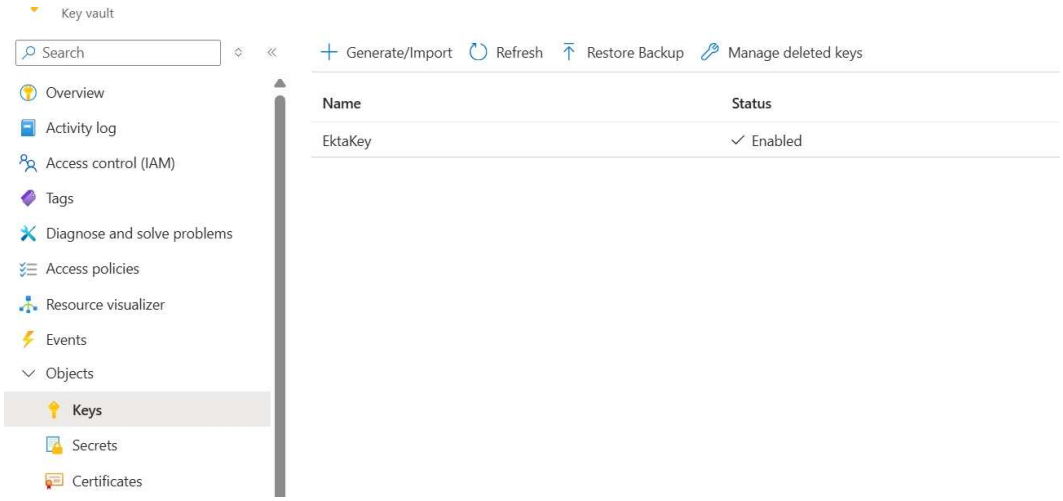
Help
Add or remove favorites by pressing Ctrl+Shift+F
Delete Move Refresh Open in mobile
Essentials
JSON View
Resource group (move) : Rg-1
Location : East US
Subscription (move) : Azure subscription 1
Subscription ID : 10c97e07-3b95-42b3-a3ef-1f506b99b039
Vault URI : https://ektavault.vault.azure.net/
Sku (Pricing tier) : Standard
Directory ID : b2fe0d10-95da-4cc1-be3f-9bd58210f4a1
Directory Name : Default Directory
Soft-delete : Enabled
Purge protection : Disabled
Tags (edit) : Add tags
Get started Properties Monitoring Tools + SDKs Tutorials
Manage keys and secrets used by apps and services
Our recommendation is to use a vault per application per environment (Development, Pre-Production and Production). This helps you to not share secrets across environments and also reduces the threat in case of a breach.
Control access to key vault
Assign access policy and
Enable logging and set up alerts
Enable logging to monitor how,
Turn on recovery options
For protection against accidental


Key vault
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Enabled

EktaVault | Secrets

Key vault

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Generate/Import | Refresh | Restore Backup | Manage deleted secrets | View sample code

Name	Type	Status
Ektaa		✓ Enabled